Herbivory, predation, disease

Learning objectives

1. Explain the relationship between growth and defense and know common defense strategies
2. Understand the role of natural enemies for maintaining diversity
3. Model an herbivore attacking seeds

last lecture: maintaining diversity, density dependence lecture mentioned natural enemies

Plant defense against natural enemies

life history theory - talked about tradeoffs plants make between growth, stasis, reproduction because of limited energy

same with defense, allocate energy to maintenance functions, growth or defense

- growth-defense tradeoff
  - grow fast: rapid germination, low defense, higher herbivory *or predation or disease*
  - well-protected: slow growth, high defense, lower herbivory
- dispersal-defense tradeoff
  - diffuse seed dispersal: hard to locate, fewer defenses
  - seed defense: aggregated seeds near parent, more defenses
- attract animals *for dispersal*, deter predators *in species that require animal seed dispersers*

Defense strategies

- physical: leaf toughness, spines, latex, hard seeds
- chemical: secondary metabolites *bitter, spicy, etc*, delayed ripening, masting *local satiation*

Natural enemies in a community

- 1 herbivore, 2 plants: apparent competition
  - growth-defense tradeoff, one plant is strong competitor, other is well-defended
  - herbivore eats both ("apparent competiton")
  - stronger competitor suffers more
- 2 herbivores, 2 plants: host-specific natural enemies, Janzen-Connell hypothesis
  - reduce seedling survival in dense areas

draw two tree species, high density of recruits around trees, enemies lead to thinning

Natural enemies in different life stages

create table and add 1 example then ask for other examples

- seed in seed bank
  - seed predators - soil invertebrates & rodents
  - pathogens
- seedling
  - insect herbivores - beetles, aphids
  - pathogens
- reproductive adults

  - insect herbivores
  - grazing mammals
  - pathogens

Modeling plant-herbivore dynamics

- model plant & herbivore pops
- plants: seedbank + new seeds - new seeds consumed
- herbivores: existing herbivores + new herbivores
- P_t - new seeds

$$P_t = \frac{fgN_t}{1 + \alpha g N_t}$$

- seed consumption:
  - C_t - seeds consumed
  - a - attack rate
  - h - handling time *rate that herbivore finds and processes a plant before next plant*
  - H_t - herbivore population

$$C_t = \frac{aP_tH_t}{1 + ahP_t}$$

Full plant model:

$$N_{t+1} = s(1-g)N_t + P_t - C_t$$

- herbivore population:
  - s_H - herbivore survival
  - b - conversion efficiency *rate that consumed plants converted into new herbivores*

$$H_{t+1} = s_H H_t + bC_t$$

Functional responses?